

Date	July 1, 2026
Team ID	SB-AI-2026-9284
Project Name	Sustainable Waste Management AI Assistant
Maximum Marks	2 Marks

Data Flow Diagram

1. Context Level DFD (Level 0)

The context level data flow maps communication vectors between the client application, Flask middleware, and third-party APIs.

- User -> Inputs Waste Item -> React Web Application
- React Client -> POST Request -> Flask Backend Controller
- Flask Backend -> Query String -> Groq AI API
- Groq AI -> JSON Schema Response -> Flask Backend
- Flask Backend -> Write Document -> Firestore Database

2. Processing DFD (Level 1)

1. User enters data or drops image.
2. Middleware triggers API classification.
3. Database Service records telemetry and increments stats collections.
4. React UI re-fetches stats API and updates state indices for Chart.js.